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COMPUTER-VISION-BASED OFFSIDE CANDIDATE DETECTION IN FOOTBALL VIDEO

DIGITAL IMAGE PROCESSING PROJECT
ELECTRONICS AND TELECOMMUNICATION ENGINEERING

HO CHI MINH CITY, 2026

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Ho Chi Minh City, date 13 month 10 year 2026

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DECLARATION OF AUTHORSHIP

I hereby declare that this thesis was carried out by myself under the guidance and supervision of Dr. Le Anh Vu; and that the work and the results contained in it are original and have not been submitted anywhere for any previous purposes. The data and figures presented in this thesis are for analysis, comments, and evaluations from various resources by my own work and have been duly acknowledged in the reference part.

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COMPUTER-VISION-BASED OFFSIDE CANDIDATE DETECTION IN FOOTBALL VIDEO

ABSTRACT

Video analysis can support the inspection of potentially offside players by combining visual detection, team identification, and spatial comparison. This study describes a football video prototype that integrates the You Only Look Once version 8 (YOLOv8) detector, Simple Online and Realtime Tracking with a deep association metric (DeepSORT), jersey color clustering, and temporal filtering of an image coordinate defensive reference. The method is presented alongside a projective formulation that explains how pitch geometry and body landmarks should be incorporated when extending the prototype beyond a fixed view approximation. Particular attention is given to the distinction between a positional candidate and an offside offence, the selection of the second last opponent, and the limitations of horizontal image coordinates under perspective projection. The available evidence consists of illustrative outputs associated with a video identified as football1.mp4. It does not include annotated test events, calibrated position measurements, or processing time benchmarks. The examples demonstrate the visualization workflow while also revealing team label inconsistencies and the need for explicit camera calibration. Consequently, the system is treated as a preliminary tool for visual review rather than a validated officiating system. A reproducible evaluation protocol is outlined to guide subsequent development.

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LIST OF ABBREVIATIONS

ATK	Attacking Group
BGR	Blue, Green, and Red
DeepSORT	Simple Online and Realtime Tracking with a Deep Association Metric
DEF	Defending Group
FN	False Negatives
FP	False Positives
FPN	Feature Pyramid Network
FPS	Frames per Second
HSV	Hue, Saturation, and Value
IFAB	International Football Association Board
OpenCV	Open Source Computer Vision Library
RANSAC	Random Sample Consensus
R CNN	Region Based Convolutional Neural Network
ResNet50	Residual Network with 50 Layers
RGB	Red, Green, and Blue
TP	True Positives
VAR	Video Assistant Referee
YOLO	You Only Look Once
YOLOv8	You Only Look Once Version 8

CHAPTER 1. INTRODUCTION

1.1 Background and motivation

The interpretation of an offside situation in association football requires the spatial relationships among attackers, opponents, and the ball to be evaluated at a particular moment of play. Broadcast video is a widely available visual source for such analysis, although perspective projection, partial occlusion, changing camera viewpoints, and the limited visibility of the ball complicate the recovery of positions on the playing field. A computer vision application can assist human inspection by locating participants, maintaining their apparent identities across successive frames, and displaying a positional reference that facilitates the examination of potentially relevant situations. This project investigates an application prototype that combines person detection, appearance based team grouping, temporal association, and filtering of a defensive reference derived from image coordinates. The resulting display provides candidate visualizations for review rather than a verified determination of an offside offence under Law 11 (1).

1.2 Objectives and scope

The central objective is to describe the application architecture and mathematical methods underlying the prototype, interpret the available visual outputs, and identify the additional evidence needed to evaluate positional accuracy. Particular attention is given to the relationship between object localization and team grouping, the selection of the second last opponent, the stabilization of a displayed reference, and the geometric restrictions imposed by an uncalibrated camera view. A further objective is to formulate a calibrated spatial extension without presenting its proposed components as completed implementation. The supplied material comprises conceptual illustrations and application screenshots but does not provide event level annotations, verified timing measurements, or a reproducible quantitative test set. Consequently, the present study is a methodological and qualitative assessment of an application rather than a numerical validation of automated offside decisions.

CHAPTER 2. THEORETICAL FOUNDATIONS AND RELATED WORK

2.1 Visual analysis of football scenes

A video frame is represented as a color array

$$I_t \in \mathbb{R}^{h_t \times w_t \times 3}, \tag{2.1}$$

where t denotes the frame index and h_t, w_t are the frame height and width. Figure 2.1 illustrates a general visual interpretation framework in which image acquisition and enhancement precede region analysis, feature representation, and object interpretation. Its knowledge base denotes contextual information that may guide several processing stages rather than an independently verified software module in the present application. In football video, frame acquisition corresponds to decoding recorded images, preprocessing provides a consistent resolution and color representation, and region analysis may restrict attention to the playing surface. Person detection, temporal association, and jersey appearance descriptors provide observations used by later positional reasoning. This dependency is important because an error in a bounding box or in team membership can propagate to the selection of the defensive reference. Generic image region encodings are not required components of the reported application.

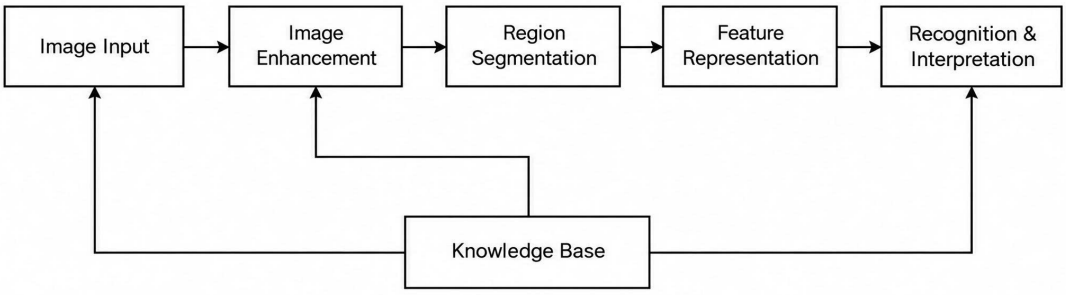


Figure 2.1. General framework for visual information processing. The knowledge base represents contextual support across stages and does not imply that each illustrated connection is implemented in the prototype.

2.2 Existing detection, tracking, and geometry methods

The You Only Look Once (YOLO) detection family provides a unified approach to object localization and classification (2). Its version 8 implementation (YOLOv8) supplies the person detector used in the reported prototype (3). Simple Online and Realtime Tracking with a deep association metric (DeepSORT) combines motion information with appearance descriptors to maintain identities across frames (4). Neither component independently identifies a player's team or establishes an offside condition. Jersey color features offer a simple basis for separating two teams. A clustering model with two centroids is convenient, but its cluster indices have no intrinsic football meaning. The connection between a cluster, a team, and an attacking direction requires additional information. The clustering objective follows the established squared error formulation of centroid clustering (5; 6). Projective methods provide complementary geometric information. Desai (7) discusses a single camera approach based on pitch line detection, a vanishing point, and estimated body keypoints. This source is an arXiv preprint and is cited as related work rather than independent validation. Its examples are not treated as experiments conducted for the present prototype. Camera calibration research by Falaleev and Chen (8) further illustrates the importance of reliable pitch correspondences and structural constraints. These studies motivate a distinction between image based visualization and calibrated spatial reasoning.

2.3 Offside position and offside offence

Under Law 11 of the International Football Association Board (IFAB), an offside position concerns eligible parts of the head, body, or feet in the opponents' half and nearer the opponents' goal line than both the ball and the second last opponent. Hands and arms are excluded. The goalkeeper participates in opponent ordering and is not removed automatically. Being level does not satisfy the positional condition (1). Position is assessed at the relevant teammate contact with the ball, normally the first contact. A goalkeeper's throw uses release. A positional condition becomes an offence only when the player becomes involved in active play under the law. Direct receipt from a goal kick, throw in, or corner kick is exempt (1). The prototype does not establish all these conditions. Accordingly, its alerts are interpreted as *offside candidates*.

2.4 Projective geometry for a calibrated extension

The reported implementation compares image coordinates obtained from bounding boxes. The following formulation specifies a calibrated extension and its assumptions. The supplied outputs do not establish that this extension has been implemented.

2.4.1. Image coordinates and pitch coordinates

The extension requires a mapping from image observations to positions on the playing surface. Let X be the longitudinal pitch coordinate, increasing toward one fixed goal, and Y the lateral coordinate. For a point on the pitch plane, $\boldsymbol{P} = (X, Y, 1)^{\mathsf{T}}$, a planar projective mapping is

$$\lambda \boldsymbol{p} = H_t \boldsymbol{P}, \qquad \boldsymbol{p} = (u, v, 1)^{\mathsf{T}}, \tag{2.2}$$

where H_t is a nonsingular 3×3 homography defined up to scale (9). With $G_t = H_t^{-1} = [g_{ab,t}]$,

$$X = \frac{g_{11,t}u + g_{12,t}v + g_{13,t}}{g_{31,t}u + g_{32,t}v + g_{33,t}}, \tag{2.3}$$

$$Y = \frac{g_{21,t}u + g_{22,t}v + g_{23,t}}{g_{31,t}u + g_{32,t}v + g_{33,t}}. \tag{2.4}$$

The denominator must be nonzero for a finite pitch point. Metric units require correspondences with known pitch coordinates, rather than an arbitrary image quadrilateral. At least four correspondences in a nondegenerate configuration are needed for a general planar mapping. Further observations permit residual checking. A pitch rectangle generally appears as a quadrilateral. A symmetric trapezoid is a restricted configuration, not a general camera model. Appendix A develops that special case without interpreting diagram lengths as calibrated field distances. Camera pan, zoom, or cuts change the mapping, which should be updated or re estimated when necessary. Pitch feature methods such as (8) address the correspondence problem. An unverified fixed mapping cannot be assumed valid throughout a broadcast.

2.4.2. Pitch region and line extraction

Pitch region selection can suppress stadium seating, graphics, and advertising. Color thresholds should be adapted to illumination and checked for missed grass or player regions. Mowing stripes can provide directional evidence when their

orientation is known, but they are not universal pitch markings and may run in an unsuitable direction. The geometric background extracts saturation from the hue, saturation, and value (HSV) color representation (10) as an edge input, followed by Canny detection and a Hough transform. Saturation may be useful for a specific recording, but no controlled comparison establishes general superiority over other channels. Canny processing includes smoothing, gradient estimation, nonmaximum suppression, and hysteresis thresholding (11). An image line can be written as

$$u\cos\theta + v\sin\theta = \rho, \tag{2.5}$$

where θ is the normal angle and ρ the signed offset (12). An angle computed from segment endpoints describes the line direction instead. The conventions must not be mixed. The original 18° to 89° orientation restriction is a view specific heuristic. The homogeneous line form,

$$\boldsymbol{\ell} = (a, b, c)^{\mathsf{T}}, \qquad \boldsymbol{\ell}^{\mathsf{T}}\boldsymbol{p} = au + bv + c = 0, \tag{2.6}$$

represents vertical lines without infinite slope division. A valid vertical line need not be rejected solely because a slope calculation is unstable.

2.4.3. Vanishing point estimation by line consensus

Lines corresponding to one family of parallel pitch lines share a vanishing point. For two distinct detected lines, a hypothesis is

$$\boldsymbol{v}^{(h)} = \boldsymbol{\ell}_j \times \boldsymbol{\ell}_k. \tag{2.7}$$

For a finite point, normalize its third coordinate to obtain $\boldsymbol{v}^{(h)} = (v_u^{(h)}, v_v^{(h)}, 1)^{\mathsf{T}}$. An incidence residual is

$$r_m(\boldsymbol{v}^{(h)}) = \frac{|\boldsymbol{\ell}_m^{\mathsf{T}}\boldsymbol{v}^{(h)}|}{\sqrt{a_m^2 + b_m^2}}, \tag{2.8}$$

and the consensus set is

$$\mathcal{I}^{(h)} = \{m : r_m(\boldsymbol{v}^{(h)}) < \tau_v\}. \tag{2.9}$$

Random Sample Consensus (RANSAC) samples line pairs, scores support, and refines a supported hypothesis (13). For a finite estimate, one refinement is

$$(\widehat{v}_u, \widehat{v}_v) = \arg\min_{v_u, v_v} \sum_{m \in \mathcal{I}^\star} w_m \frac{(a_m v_u + b_m v_v + c_m)^2}{a_m^2 + b_m^2}, \tag{2.10}$$

where $w_m \geq 0$ may encode segment support. The fitted model is a point supported by incident lines. Fitting an arbitrary line through pairwise intersection points does not, by itself, estimate that model. Near parallel detections also produce unstable intersections. Distant or infinite vanishing points require a suitable homogeneous or angular treatment.

2.4.4. Body landmarks and ground plane validity

The anatomical landmark detector uses a region based convolutional neural network (R CNN) and is referred to here as Keypoint R CNN. Its backbone combines a residual network with 50 layers (ResNet50) and a feature pyramid network (FPN) (14; 15; 16; 17).

The documented model identifier is `keypointrcnn_resnet50_fpn`. Write its landmarks as

$$\mathcal{K}_{i,t} = \{(u_{i,r,t}, v_{i,r,t}, q_{i,r,t})\}_{r=1}^{R}, \tag{2.11}$$

where r denotes the anatomical landmark and $q_{i,r,t}$ its confidence. Confidence filtering and semantic landmark selection are both necessary. Landmarks represent posture more specifically than a whole box. However, a shoulder or ankle joint is not automatically the outermost relevant body surface. A limited joint set may miss a leaning head or projecting toe. A pitch plane homography applies to points on that plane. Mapping an elevated shoulder or knee through its inverse gives the viewing ray intersection with the ground, not the true vertical projection of that body point. Connecting every elevated keypoint directly to a pitch plane vanishing point therefore does not resolve body height ambiguity. A calibrated camera with a suitable three dimensional model, or additional views, is required when that ambiguity matters. Ground contact approximations should be labeled accordingly. These restrictions follow from the plane assumption of a homography and the separation of ground position from object height (9).

2.4.5. Perspective consistent reference lines

For $X = X_{\mathrm{ref},t}$, the pitch plane line is

$$\boldsymbol{L}_t = (1, 0, -X_{\mathrm{ref},t})^{\mathsf{T}}. \tag{2.12}$$

Its image is

$$\boldsymbol{\ell}_t \propto H_t^{-\mathsf{T}} \boldsymbol{L}_t. \tag{2.13}$$

Indeed, for $\boldsymbol{p} \propto H_t \boldsymbol{P}$,

$$(H_t^{-\mathsf{T}} \boldsymbol{L}_t)^{\mathsf{T}} H_t \boldsymbol{P} = \boldsymbol{L}_t^{\mathsf{T}} \boldsymbol{P} = 0.$$

The resulting image line is generally oblique. This transformation follows the dual point and line representation of planar projective geometry (9). The lateral pitch direction has homogeneous point $\boldsymbol{d}_Y = (0, 1, 0)^{\mathsf{T}}$, with vanishing point

$$\boldsymbol{v}_Y \propto H_t \boldsymbol{d}_Y. \tag{2.14}$$

For a valid ground plane reference point, the same line can be formed as

$$\boldsymbol{\ell}_t \propto \boldsymbol{v}_Y \times \boldsymbol{p}_{\mathrm{ref}}. \tag{2.15}$$

This requires the vanishing point of lines parallel to the goal line, not an arbitrary family of grass edges. A vanishing point alone does not supply full metric calibration (9).

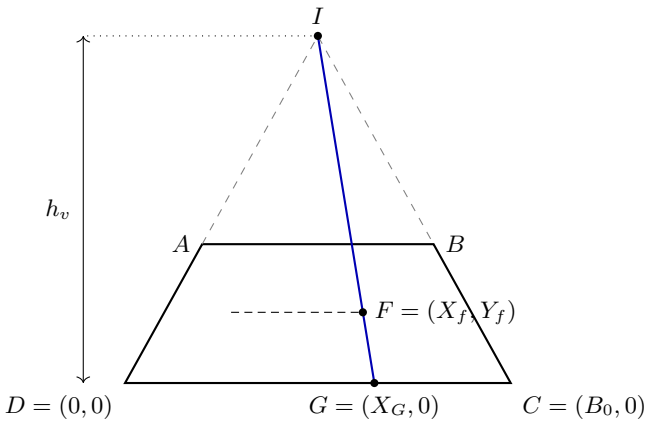


Figure 2.2. Editable schematic for Appendix A. Diagram coordinates are not field measurements.

2.4.6. Reference positional criterion in pitch coordinates

Let pitch length be L, and define

$$s(X) = \begin{cases} X, & \text{attack toward } X = L, \\ L - X, & \text{attack toward } X = 0. \end{cases} \tag{2.16}$$

This coordinate increases toward the attacked goal. Let $s_i^{\text{elig}}(t)$ denote the furthest eligible body position. It is not a box extremity unless that approximation is justified. Sort opponent values in descending order, $s_{[1]}^{\text{elig}}(t) \geq s_{[2]}^{\text{elig}}(t) \geq \cdots$, and let $s_{\text{B}}(t)$ represent the ball reference. The longitudinal threshold is

$$s_{\text{ref}}(t) = \max\{s_{[2]}^{\text{elig}}(t), s_{\text{B}}(t)\}. \tag{2.17}$$

A screening indicator at the relevant event time $t^\star$ is

$$C_i(t^\star) = \mathbf{1}\left\{ s_i^{\text{elig}}(t^\star) > \frac{L}{2} \text{ and } s_i^{\text{elig}}(t^\star) > s_{\text{ref}}(t^\star) \right\}. \tag{2.18}$$

This expression formalizes the positional component summarized in Section 2.3 (1). It does not determine involvement in active play or other event conditions. Uncertainty should be handled separately. For $\widehat{\Delta}_i = \widehat{s}_i^{\text{elig}} - \widehat{s}_{\text{ref}}$ with a justified bound $e_i \geq 0$,

$$\widehat{\Delta}_i - e_i > 0 \quad \text{indicates confidently beyond the reference,} \tag{2.19}$$

$$\widehat{\Delta}_i + e_i < 0 \quad \text{indicates confidently behind the reference.} \tag{2.20}$$

The intervening interval is unresolved. The bound must reflect calibration, landmark, and timing errors. A fixed pixel threshold is not automatically a metric uncertainty bound.

CHAPTER 3. SYSTEM DESIGN AND IMPLEMENTATION

3.1 Processing environment and documented configuration

The reported environment consists of Python on Windows, with PyCharm used for development. The Open Source Computer Vision Library (OpenCV) handles video decoding, image conversion, geometric drawing, and display (10; 18). NumPy supports numerical operations, Ultralytics supplies the detector, the Python clustering library supplies centroid clustering, and FilterPy supplies filtering routines (3; 6; 19). DeepSORT is described as the temporal association component. A separate model for detecting anatomical keypoints appears in the geometric background, but its integration into the final video pipeline is not established by the available material. Table 3.1 separates documented settings from information needed for reproduction. The reported execution was not independently rerun for this document.

Table 3.1. Documented prototype configuration and reproducibility limits.

Item	Available information
Input	Local video identified as `football1.mp4`. Duration, frame rate, provenance, and event annotations are unspecified in the supplied report.
Frame size	Frames are described as resized to 1480×820 pixels.
Detection	YOLOv8 person detections. The checkpoint, confidence threshold, and minimum box height are unspecified in the report.
Tracking	DeepSORT is reported. Association thresholds and track management settings are unspecified.
Team grouping	Clustering with two centroids using mean color features.
Color center update	Previous center weight 0.8, current center weight 0.2.
Reference smoothing	Kalman filtering and a recent position buffer. Final median and mean weights are 0.7 and 0.3.
Unreported settings	Box edge weight, candidate tolerance, history length, hold duration, covariance matrices, software versions, and hardware specifications.

3.2 Processing architecture

Figure 3.1 summarizes the processing stages reported for the application. Video frames provide detections that are associated over time and grouped by jersey appearance. A heuristic assigns team roles and attack direction, after which the observed opponents are ordered to construct an image coordinate reference. This reference is filtered and used to display positional candidates. The pipeline

is intended for screening and visualization, and the calibrated formulation in Section 2.4 remains a proposed geometric extension rather than evidence of an implemented field calibration.

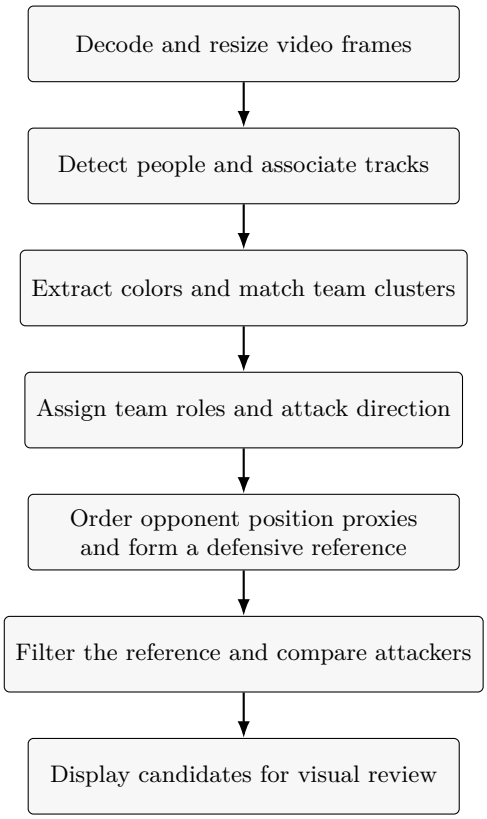


Figure 3.1. Reported image coordinate prototype workflow. Team role assignment and opponent visibility are prerequisites for interpreting the defensive reference.

The more detailed frame processing logic is presented in Fig. 3.2. It makes explicit the need to handle frames with too few detections or unreliable team assignments before attempting a positional comparison. The illustrated minimum of four players is a diagrammatic processing gate rather than a sufficient condition for correct opponent ordering, which also depends on whether the relevant opponents are actually visible. The final decision node identifies a potential positional candidate and does not test the ball contact instant or involvement in active play. The flowchart is therefore interpreted as an application level decision design consistent with a visual screening task. Its individual branching thresholds have not been verified against source code or annotated recordings.

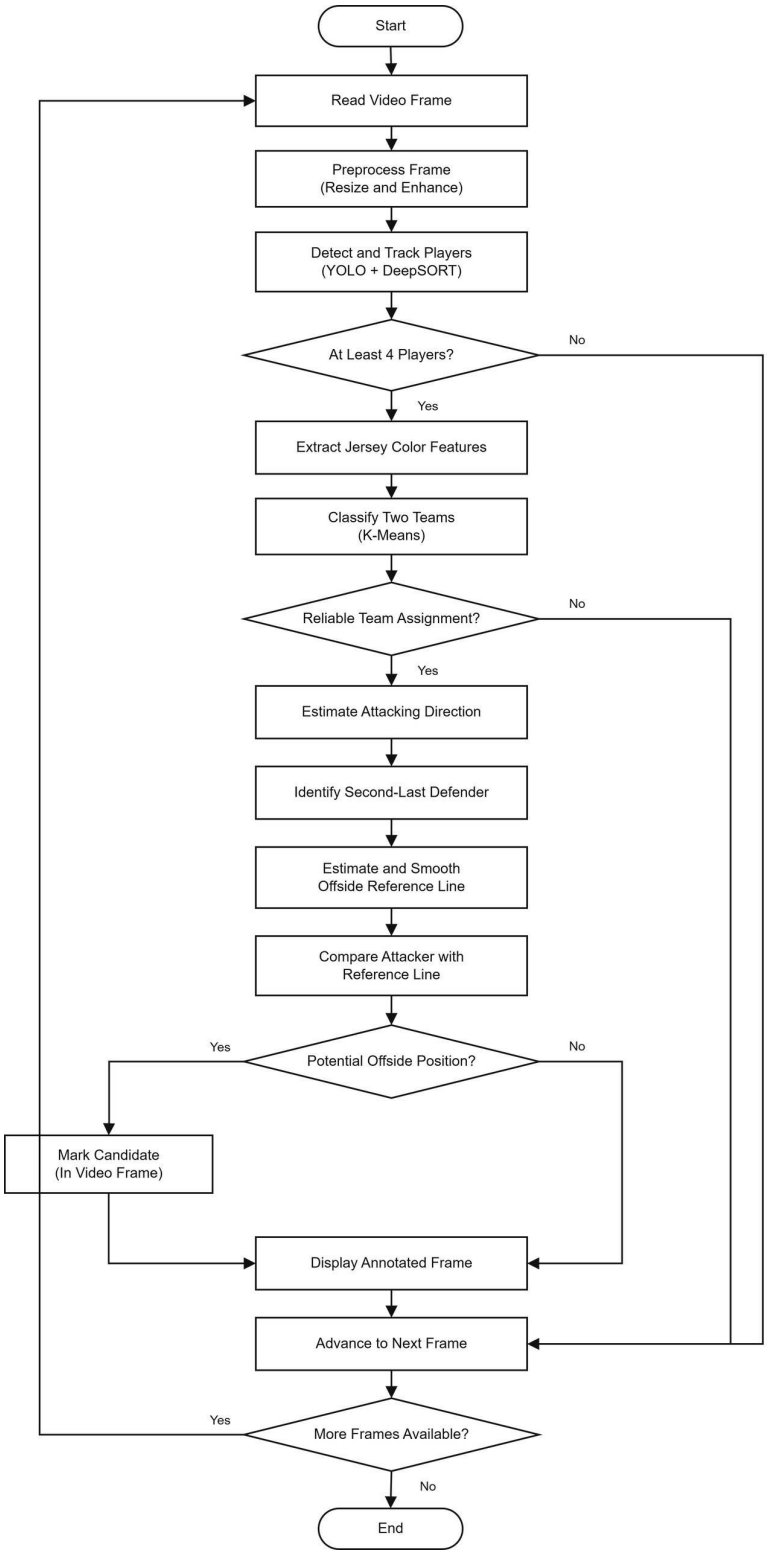


Figure 3.2. Detailed frame processing flow for candidate visualization. The player count gate, assignment reliability check, and potential position test are design stage conditions and do not constitute a validated offside decision.

3.3 Video acquisition, detection, and tracking

3.3.1. Frame acquisition and preprocessing

OpenCV reads the input video sequentially. Resizing each frame to a common display size simplifies downstream thresholds and drawing coordinates. However, resizing alone does not establish a geometric calibration or guarantee real time execution. For an original resolution $W_0 \times H_0$, direct resizing uses

$$u' = s_x u, \quad v' = s_y v, \qquad s_x = \frac{1480}{W_0}, \quad s_y = \frac{820}{H_0}. \tag{3.1}$$

Coordinates and calibration matrices must use this resized frame, including offsets if padding preserves the aspect ratio. The reported minimum height and region filters have unspecified thresholds. Their evaluation must account for small or distant players whose omission could alter opponent ordering.

3.3.2. Person detection

For frame t, the detector returns

$$\mathcal{D}_t = \{(\boldsymbol{b}_{i,t}, c_{i,t}, q_{i,t})\}_{i=1}^{n_t}, \qquad \boldsymbol{b}_{i,t} = (u_{i,t}^{\min}, v_{i,t}^{\min}, u_{i,t}^{\max}, v_{i,t}^{\max}), \tag{3.2}$$

where $c_{i,t}$ is a class label and $q_{i,t}$ a confidence score. The prototype retains person detections subject to confidence and size filtering (3). A person label is not equivalent to a player label: referees, staff, or spectators can also enter this set. Boxes provide image regions for appearance extraction and approximate positional analysis. They do not isolate the body parts relevant to offside. A box extremity can be determined by an arm, a loose garment, or background pixels.

3.3.3. Temporal association using DeepSORT

DeepSORT associates detections with persistent identities using motion prediction and appearance information (4). A track identifier allows the same person to be referenced across frames, supporting color label consistency and candidate display persistence. Track association is useful during short occlusions, but similar jerseys, crowded scenes, abrupt camera motion, and long occlusions can produce fragmented tracks or identity switches. Predicted boxes without a current supporting detection should be distinguished from observed boxes. The filter used internally by a tracker also serves a different purpose from the separate

filter stabilizing the displayed defensive reference.

3.4 Appearance grouping and attack direction

3.4.1. Jersey color features and team clustering

For selected pixels $\Omega_{i,t}$ from detection i, the reported mean color representation is

$$\boldsymbol{c}_{i,t} = \frac{1}{|\Omega_{i,t}|} \sum_{(u,v)\in\Omega_{i,t}} \begin{bmatrix} R_t(u,v) \\ G_t(u,v) \\ B_t(u,v) \end{bmatrix}. \tag{3.3}$$

The red, green, and blue (RGB) feature order must match the array passed to the feature extractor. OpenCV commonly stores these channels in blue, green, and red (BGR) order (10). A whole box mean is not necessarily the dominant jersey color. Grass, skin, shorts, and overlapping people may influence it. For two clusters, the grouping objective is

$$\min_{\{\mathcal{C}_{k,t},\boldsymbol{\mu}_{k,t}\}_{k=1}^{2}} \sum_{k=1}^{2} \sum_{i\in\mathcal{C}_{k,t}} \|\boldsymbol{c}_{i,t} - \boldsymbol{\mu}_{k,t}\|_2^2, \tag{3.4}$$

where $\mathcal{C}_{k,t}$ and $\boldsymbol{\mu}_{k,t}$ denote a cluster and its center (5; 6). The labels are arbitrary and may exchange between frames. A consistent two center matching rule is

$$\pi_t^\star = \arg\min_{\pi\in\mathfrak{S}_2} \sum_{k=1}^{2} \|\boldsymbol{\mu}_{\pi(k),t}^{\text{cur}} - \boldsymbol{\mu}_{k,t-1}^{\text{ref}}\|_2^2, \tag{3.5}$$

where $\mathfrak{S}_2$ contains the two label permutations. After correspondence is established, the documented update becomes

$$\boldsymbol{\mu}_{k,t}^{\text{ref}} = 0.8\,\boldsymbol{\mu}_{k,t-1}^{\text{ref}} + 0.2\,\boldsymbol{\mu}_{\pi_t^\star(k),t}^{\text{cur}}. \tag{3.6}$$

Equation (3.5) formalizes the reported correspondence principle, whose exact code realization remains unverified. Two clusters cannot represent teams, officials, and distinct goalkeeper kits. Torso crops and an explicit outlier or role classifier offer development options, while goalkeepers must remain in the correct opponent set regardless of their clothing.

3.4.2. Team roles and attack direction heuristic

The report estimates attack direction by comparing the groups' mean horizontal positions:

$$\bar{u}_{\mathrm{A},t} = \frac{1}{|\mathcal{A}_t|} \sum_{i \in \mathcal{A}_t} u_{i,t}, \qquad \bar{u}_{\mathrm{D},t} = \frac{1}{|\mathcal{O}_t|} \sum_{j \in \mathcal{O}_t} u_{j,t}, \tag{3.7}$$

where $u_{i,t}$ is a box position proxy, $\mathcal{A}_t$ the candidate attacking set, and $\mathcal{O}_t$ the opponent set. The heuristic assigns rightward motion when $\bar{u}_{\mathrm{A},t} > \bar{u}_{\mathrm{D},t}$, and leftward motion otherwise. The centroid comparison is a heuristic. Counterattacks and camera pans can shift team centroids without changing the target goals, while color clusters alone do not determine possession. A more reliable extension would initialize attack direction from match context or calibrated pitch orientation and update it when the playing ends change.

3.5 Defensive reference and temporal stabilization

3.5.1. Image coordinate defensive reference

The reported position approximation is a weighted horizontal box coordinate,

$$u_{i,t}^{\mathrm{edge}} = \alpha u_{i,t}^{\min} + (1-\alpha) u_{i,t}^{\max}, \qquad 0 \le \alpha \le 1. \tag{3.8}$$

The description does not specify α or its dependence on attack direction. Since an interior weighted point is not generally the leading body boundary, it is more accurately termed a *position proxy*. Let $d_t = +1$ denote an attack toward increasing image u, and $d_t = -1$ the opposite convention. Define

$$s_{i,t}^{\mathrm{img}} = d_t u_{i,t}^{\mathrm{edge}}. \tag{3.9}$$

For the observed opponents, sort these values in descending order,

$$s_{[1],t}^{\mathrm{img}} \ge s_{[2],t}^{\mathrm{img}} \ge \cdots . \tag{3.10}$$

The second last opponent proxy is

$$u_t^{\mathrm{ref}} = d_t s_{[2],t}^{\mathrm{img}}. \tag{3.11}$$

Equivalently, sorting ordinary u values in ascending order selects the second largest value for a rightward attack and the second smallest for a leftward attack.

At least two correctly identified opponents must be present. This formulation resolves ambiguity in references to the last defender. The opponent set includes the goalkeeper, and visibility of two opponents does not establish that all opponents nearer the goal have been observed. Image u ordering is only a view dependent approximation to pitch longitudinal ordering.

3.5.2. Kalman filtering and recent history smoothing

The prototype describes Kalman filtering of the reference and further smoothing with a short buffer. A standard constant velocity specification is

$$\boldsymbol{x}_t = (u_t, \dot{u}_t)^{\mathsf{T}}, \quad \boldsymbol{x}_t = F_t \boldsymbol{x}_{t-1} + \boldsymbol{w}_t, \quad z_t = H\boldsymbol{x}_t + v_t, \tag{3.12}$$

where

$$F_t = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, \qquad H = \begin{bmatrix} 1 & 0 \end{bmatrix}. \tag{3.13}$$

Here $z_t = u_t^{\mathrm{ref}}$, while Q_t, R_t are the covariances of process noise and measurement noise. This is a conventional specification of the reported idea. The exact matrices used in code are not available. The prediction and correction equations are (20)

$$\widehat{\boldsymbol{x}}_{t|t-1} = F_t \widehat{\boldsymbol{x}}_{t-1|t-1}, \tag{3.14}$$

$$P_{t|t-1} = F_t P_{t-1|t-1} F_t^{\mathsf{T}} + Q_t, \tag{3.15}$$

$$K_t = P_{t|t-1} H^{\mathsf{T}} (H P_{t|t-1} H^{\mathsf{T}} + R_t)^{-1}, \tag{3.16}$$

$$\widehat{\boldsymbol{x}}_{t|t} = \widehat{\boldsymbol{x}}_{t|t-1} + K_t (z_t - H\widehat{\boldsymbol{x}}_{t|t-1}), \tag{3.17}$$

$$P_{t|t} = (I - K_t H) P_{t|t-1}. \tag{3.18}$$

For a recent history $\mathcal{B}_t$, the documented display combination is

$$u_t^{\mathrm{display}} = 0.7 \ \mathrm{median}(\mathcal{B}_t) + \frac{0.3}{|\mathcal{B}_t|} \sum_{b \in \mathcal{B}_t} b. \tag{3.19}$$

The buffer length and its placement relative to the Kalman update are unspecified. Figure 3.3 provides an illustrative comparison between a fluctuating raw reference and a temporally stabilized display. It explains the intended reduction of frame to frame jitter but is not a measured before and after experiment from the supplied video. In particular, the straight line used in the illustration does not imply that Kalman filtering rectifies perspective, recovers the exact location of the second last opponent, or makes a positional decision more accurate. Filtering

can introduce lag, and it can blend measurements from different players when the identity of the reference changes. Camera cuts and substantial changes of viewpoint require explicit state management. Therefore, positional measurements at the relevant event frame must be evaluated separately from the historical estimate used for display.

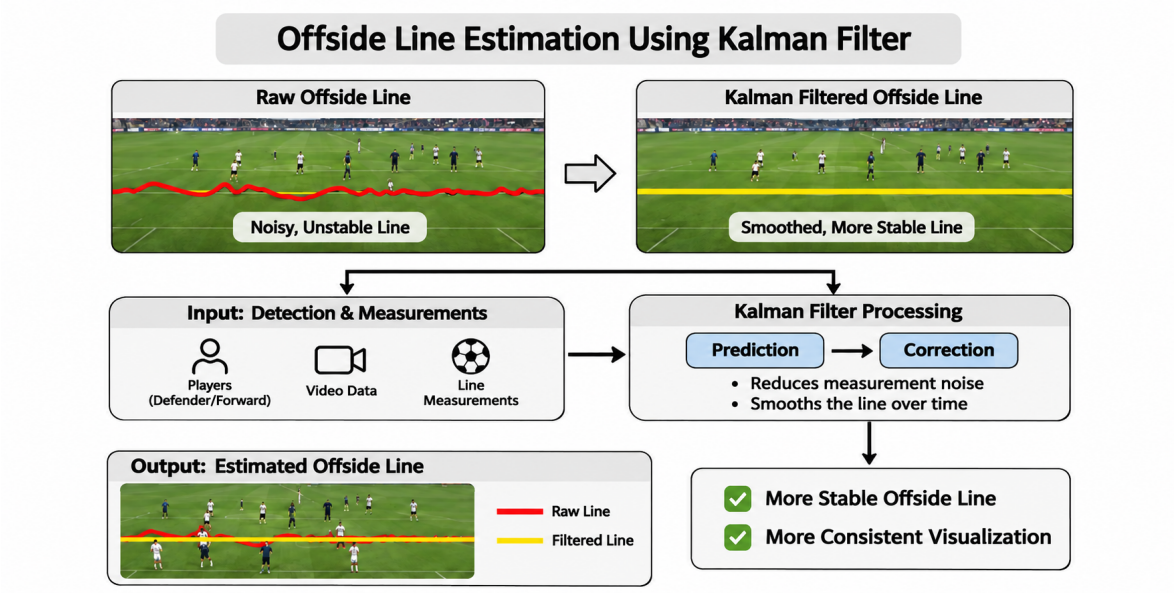


Figure 3.3. Conceptual visualization of temporal reference filtering. The smoother line illustrates reduced display fluctuation and is not quantitative evidence of geometric correctness or decision accuracy.

3.6 Candidate highlighting and display persistence

The signed image coordinate gap is

$$g_{i,t}^{\mathrm{img}} = d_t(u_{i,t}^{\mathrm{edge}} - u_t^{\mathrm{display}}). \tag{3.20}$$

An attacker may be highlighted when this gap exceeds an image tolerance τ_u within the selected attacking region. If one candidate is displayed, the selection principle is

$$i_t^\star = \arg\max_{\substack{i \in \mathcal{A}_t: \\ g_{i,t}^{\mathrm{img}} > \tau_u}} g_{i,t}^{\mathrm{img}}, \tag{3.21}$$

provided the set is nonempty. A hold counter briefly preserves the displayed identity. The full candidate set should remain available because subsequent involvement can concern another player. The alert omits independently established

ball position, teammate contact timing, and active play interpretation. Green, blue, and red boxes identify the attacking group (ATK), defending group (DEF), and candidates. OFFSIDE and VAR CHECK, referring to a video assistant referee (VAR), are interface labels (21). They do not establish an official decision or affiliation with an officiating system.

CHAPTER 4. RESULTS AND DISCUSSION

4.1 Player grouping and scope of evidence

The available evidence consists of representative application screenshots associated with the recorded football video. These outputs show the spatial placement of person detection boxes, the assigned team labels, and the reference and candidate overlays. They do not provide a sequence of ground truth identities, verified event timestamps, or measurements of detection, tracking, and decision performance. Figure 4.1 shows that light colored and dark colored kits are frequently assigned distinct labels, with the interface displaying ATK and DEF alongside the detected people. Nevertheless, a player wearing a light colored kit near the lower right of the frame appears under the DEF label. This observable inconsistency demonstrates why clustering based only on image color should be treated as an appearance grouping stage and not as conclusive evidence of team role. The screenshot establishes that the labeling and visualization components can generate a combined display, but it does not establish the correctness of every player assignment.

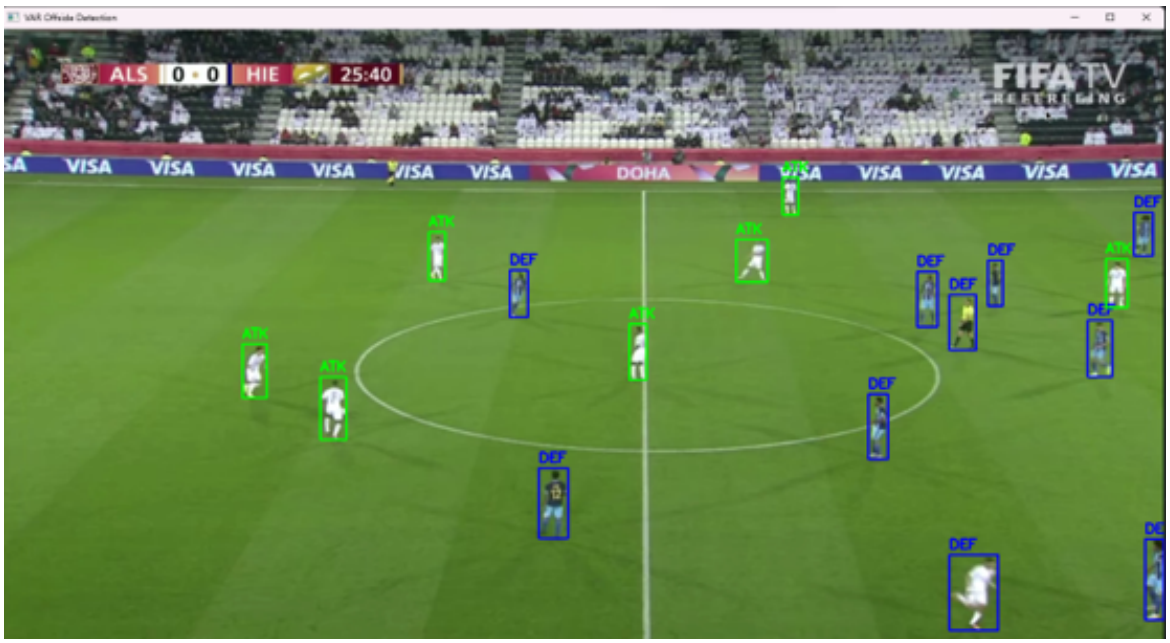


Figure 4.1. Observed player detections and team labels in the application interface. A light colored player near the lower right is assigned DEF, indicating a visible limitation of appearance based grouping.

4.2 Candidate visualization and camera perspective

Figure 4.2 illustrates a frame in which the application displays a vertical red reference together with a highlighted player and an OFFSIDE label. This output confirms that the interface combines player detections, assigned team roles, and a candidate indicator in a single visualization. The reference is drawn from $(u_t^{\text{display}}, 0)$ to $(u_t^{\text{display}}, h_t - 1)$, which is a line of constant horizontal image coordinate. Since the apparent goal line direction is influenced by projective geometry, a vertical screen line generally does not correspond to a physical line parallel to the goal line. The overlay must consequently be interpreted as an image coordinate aid rather than an automatically calibrated offside line. The screenshot alone also cannot establish the moment of teammate ball contact, the location of the ball, the identity of the second last opponent, or a subsequent offence.

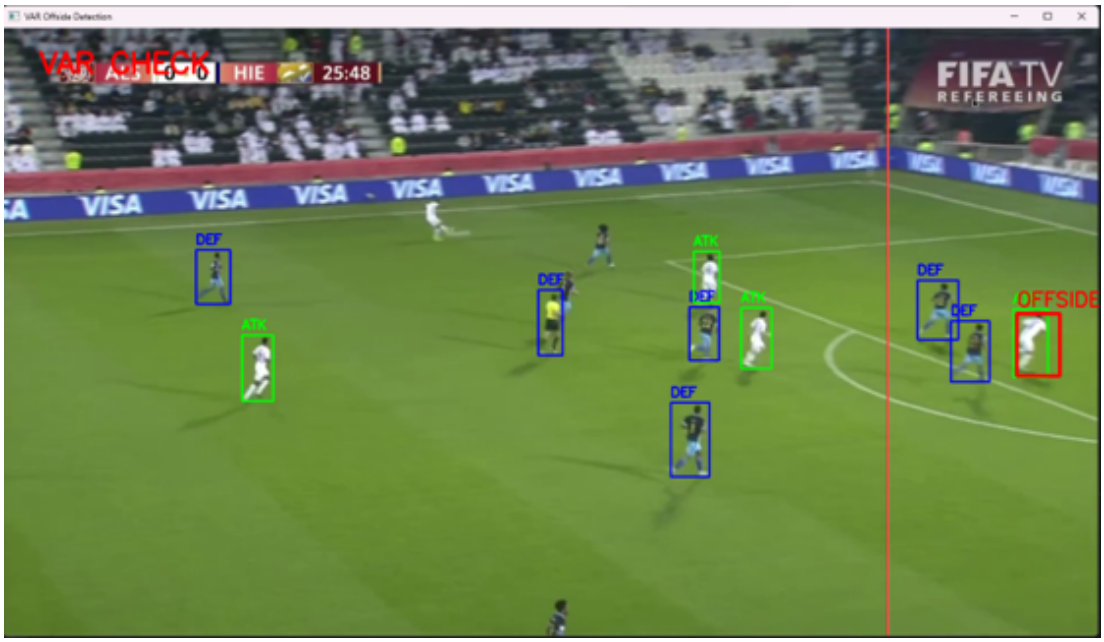
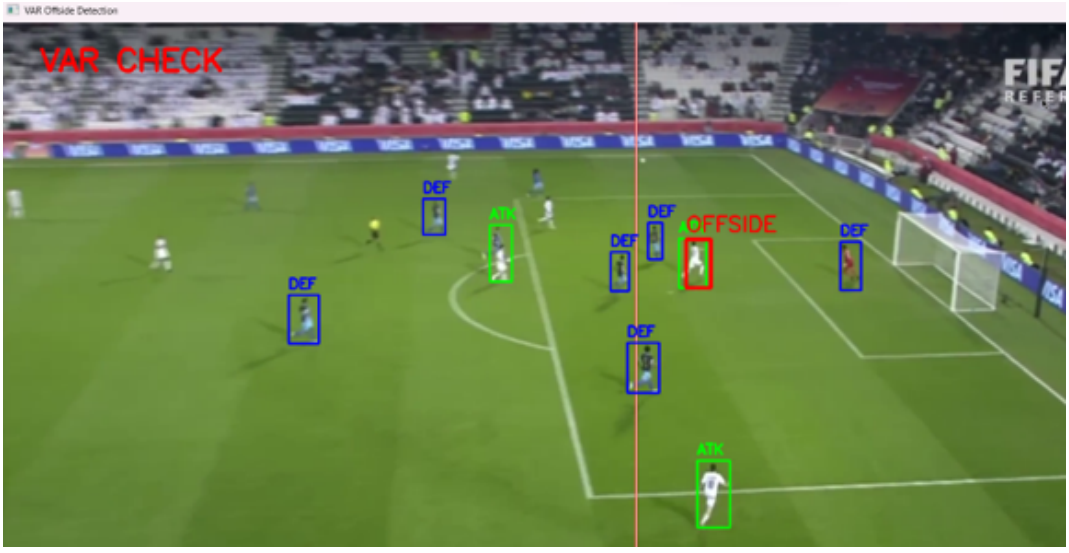


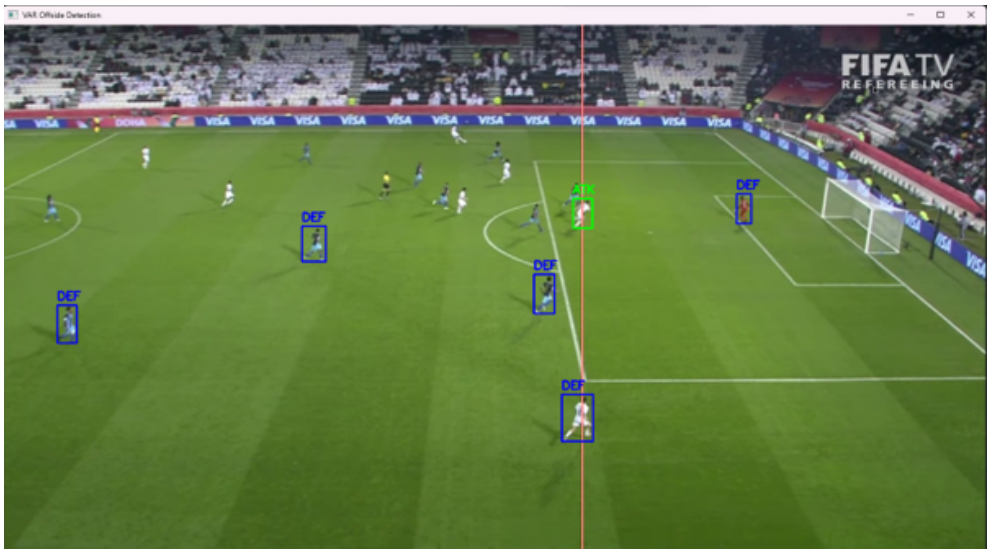
Figure 4.2. Application interface showing a positional candidate, a vertical reference, and a software generated alert. The OFFSIDE and VAR CHECK labels are not official match decisions.

Figure 4.3 provides two further views in which the player distribution and apparent scene geometry change while the software continues to show bounding boxes and an image coordinate reference. These examples demonstrate the generation of overlays for different frames, although still images do not reveal temporal identity consistency, the amount of filtering lag, or whether the

positional reference remains correct under a moving camera. Multiplying a horizontal coordinate by an attack direction sign can reverse the comparison direction, but it cannot remove the nonuniform mapping between screen pixels and pitch distances. Similarly, smoothing may improve visual continuity without reducing spatial bias. The supplied images therefore support a qualitative demonstration of application integration and expose clear sources of uncertainty, but do not justify a numerical claim about reliability or equivalence with an official video review system.



(a) Candidate visualization at one camera view.



(b) Reference visualization at a changed player distribution.

Figure 4.3. Additional frames from the application interface. These outputs document overlay generation across views and do not provide evidence of calibrated positional accuracy.

4.3 Quantitative evaluation protocol

4.3.1. Reproducible data and protocol

A subsequent quantitative study should identify video sources, resolution, frame rate, match and event counts, and the method for selecting relevant contact frames. Clips from the same match should not be split between training and testing in a way that leaks nearly identical content. Use of pretrained models without football specific training should be stated explicitly. Annotations should distinguish person boxes, teams, goalkeeper and official roles, track identities, pitch correspondences, eligible body references, ball positions, and event outcomes. Test events should be reviewed independently, with ambiguous cases retained as uncertain. Hardware, package versions, checkpoints, and all settings in Table 3.1 should also be recorded.

4.3.2. Component and event level metrics

Detection precision and recall require annotated boxes and a stated overlap criterion. Team assignment accuracy should specify how officials, goalkeepers, and rejected detections are handled. Tracking evaluation should include identity switches and fragmentation, rather than visual smoothness alone. For M matched ground plane reference points,

$$E_{\text{pos}} = \frac{1}{M} \sum_{m=1}^{M} \|\widehat{\boldsymbol{P}}_m - \boldsymbol{P}_m^{\text{gt}}\|_2, \tag{4.1}$$

where both two dimensional vectors use common physical units. A reference position error is

$$E_{\text{ref}} = \frac{1}{M} \sum_{m=1}^{M} |\widehat{X}_{\text{ref},m} - X_{\text{ref},m}^{\text{gt}}|. \tag{4.2}$$

An image line error can also be reported, but must be identified as such and evaluated at stated image locations. For candidate detection at the event level, true positives (TP) count correctly highlighted positive events, false positives (FP) count false alerts, and false negatives (FN) count missed positive events. The standard metrics are (22)

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{4.3}$$

and

$$F_1 = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}, \tag{4.4}$$

when the denominators are nonzero. Positional labels and offence labels must be evaluated separately. An abstention policy should also report its coverage so uncertain events are not silently excluded. Runtime requires complete pipeline measurements. If frame t takes T_t seconds, throughput in frames per second (FPS) over N_f frames is

$$\mathrm{FPS} = \frac{N_f}{\sum_{t=1}^{N_f} T_t}. \tag{4.5}$$

Latency distributions, hardware, and video decoding and display conventions should accompany this value. No timing data in the current material support such a calculation.

4.3.3. Ablation and error analysis

A controlled comparison should examine the raw reference, Kalman filtering alone, and Kalman filtering combined with a short history buffer. Both frame to frame jitter and the spatial error at the relevant event instant must be reported so that visual smoothness is not mistaken for improved positional accuracy. This comparison would also quantify lag under sudden player movement and camera transitions. Further comparisons should isolate whole box versus torso color features, centroid based versus initialized attack direction, and image coordinate versus calibrated pitch coordinate ordering. Performance should be stratified by occlusion, camera movement, zoom, player scale, and distance from the positional boundary. Large margin cases do not establish accuracy for marginal decisions. Missed goalkeepers, mixed role clusters, and incomplete opponent visibility should be treated as distinct error causes.

4.4 Limitations and reproducibility

The evidence has five principal limitations. Screenshots do not validate temporal tracking or filtering. Two color clusters cannot represent all football roles. Weighted box coordinates do not isolate eligible body surfaces. Horizontal image ordering and a vertical reference lack general perspective consistency. Finally, ball contact timing, the ball reference, and active play interpretation remain unverified. These limitations motivate role aware classification, complete opponent visibility

checks, dynamic calibration, body position reconstruction, and uncertainty aware review. Reproduction also requires the provenance of `football1.mp4`, relevant source files, model checkpoints, package versions, complete settings from Table 3.1, and event annotations. A public experimental package is not established by the supplied report, and the displayed results were not independently reproduced for this document.

CHAPTER 5. CONCLUSIONS AND FUTURE WORK

5.1 Conclusions

This study has presented the design, operation, and geometric assumptions of a computer vision application that highlights potential offside positions in recorded football video. The documented prototype combines person detection using YOLOv8, temporal association using DeepSORT, jersey appearance clustering, and stabilization of an image coordinate defensive reference. The included screenshots demonstrate the integration of detections and positional alerts in a common interface, while also revealing an incorrect appearance assignment and the limitations of a vertical screen reference under perspective projection. The image processing and filtering illustrations clarify methodological intent but are not experimental evidence of accuracy. The analysis identifies calibrated pitch geometry, valid opponent ordering, the independent ball reference, and suitable estimates of eligible body position as requirements for dependable positional assessment. Because annotated outcomes and runtime experiments are absent, the contribution is best characterized as an application prototype and evaluation framework rather than a validated automated offside officiating system.

5.2 Future work

The next phase should establish a reproducible experimental environment and evaluate the existing perception stages with manually checked labels. Team identification requires explicit treatment of referees, distinct goalkeeper kits, ambiguous appearance features, and incomplete opponent visibility. The geometric component should then maintain calibrated pitch correspondences under camera motion and select body references that are consistent with the playing rules. Figure 5.1 depicts a possible integration of pitch detection, player detection, ball detection, team assignment, and player identification. This diagram is included as a development architecture rather than as a screenshot of functions verified in the present prototype. In particular, the visible interface outputs do not independently confirm that pitch detection and ball detection were active in the reported pipeline.

A subsequent study should associate each candidate observation with the

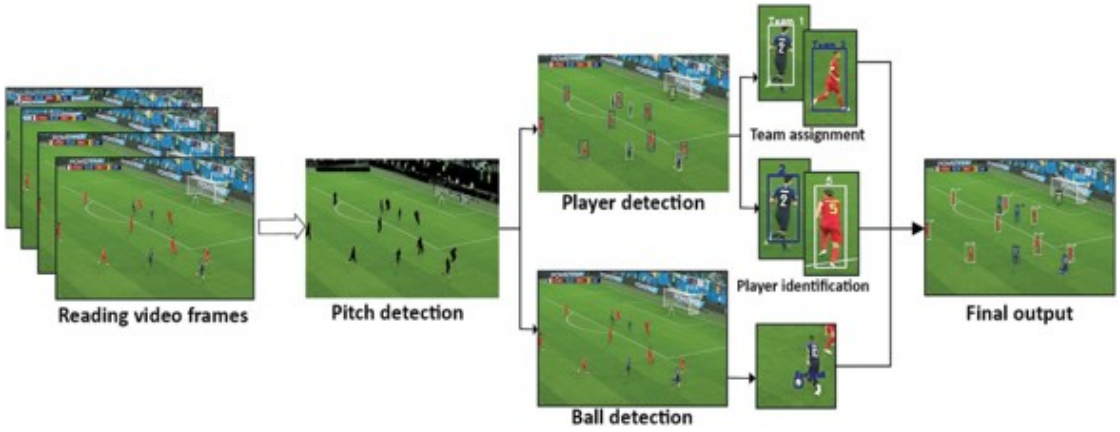


Figure 5.1. Conceptual architecture for extending the application with pitch and ball observations alongside player localization and identity association. These additional stages require implementation and independent evaluation.

appropriate teammate ball contact instant and distinguish positional eligibility from active involvement under the official law. A controlled evaluation should compare predictions with independently annotated event frames, report abstentions for unresolved situations, and measure both reference line accuracy and the processing cost on documented hardware. Errors should be separated by detection, appearance assignment, tracking, calibration, temporal synchronization, and candidate classification. These development priorities would move the application from a qualitative visualization prototype toward a testable decision support system without attributing unperformed experiments to the present work.

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CHAPTER A. IMAGE PLANE GEOMETRY AND COORDINATE INTERPRETATION

A.1 Optical magnification

The thin lens relations discussed in the theoretical material are (23)

$$\frac{1}{f} = \frac{1}{p} + \frac{1}{q}, \qquad \left|\frac{h_{\text{img}}}{h_{\text{obj}}}\right| = \left|\frac{q}{p}\right|, \tag{A.1}$$

where f is focal length, p, q are object and image distances, and $h_{\text{obj}}, h_{\text{img}}$ are physical heights. Thus

$$|q| = |p| \left|\frac{h_{\text{img}}}{h_{\text{obj}}}\right|. \tag{A.2}$$

These equations explain magnification, but do not supply unknown depth, orientation, or pitch scale. Pixel height is not physical image plane height without sensor calibration. Section 2.4 provides the operative model for planar pitch mapping.

A.2 Symmetric trapezoid construction

In Fig. 2.2, use an upward positive diagram coordinate system, distinct from downward positive video pixels. Let

$$D = (0, 0), \quad C = (B_0, 0), \quad I = (B_0/2, h_v), \quad F = (X_f, Y_f),$$

where I is a finite vanishing point, $B_0 = DC > 0$, and $h_v \neq Y_f$. Let $G = (X_G, 0)$ be the intersection of IF with the baseline. Parameterizing the line,

$$(x(\lambda), y(\lambda)) = (B_0/2, h_v) + \lambda(X_f - B_0/2, Y_f - h_v). \tag{A.3}$$

Setting its vertical coordinate to zero gives

$$0 = h_v + \lambda(Y_f - h_v) \quad \Longrightarrow \quad \lambda = \frac{h_v}{h_v - Y_f}. \tag{A.4}$$

Consequently,

$$X_G = \frac{B_0}{2} + \frac{h_v(X_f - B_0/2)}{h_v - Y_f}. \tag{A.5}$$

If the side slope in this symmetric drawing is expressed by a positive vertical increment YA and horizontal increment XA, then

$$h_v = \frac{B_0}{2}\frac{YA}{XA}, \tag{A.6}$$

and

$$X_G = \frac{B_0}{2} + \frac{\left(\frac{B_0}{2}\frac{YA}{XA}\right)\left(X_f - \frac{B_0}{2}\right)}{\left(\frac{B_0}{2}\frac{YA}{XA}\right) - Y_f}. \tag{A.7}$$

This is a baseline intersection coordinate in the same diagram, not a metric player coordinate or complete two dimensional rectification. A new origin or reversed axis must be defined before adding offsets or changing signs.

A.3 Local breadth coordinate ratios

Consider a trapezoid with horizontal AB, VT, DC, where T lies on side BC. Let K be the perpendicular projection of B onto DC, and let $N = BK \cap VT$. The height of T above DC is NK. The right triangle formed by T, C, and the perpendicular projection of T onto DC is similar to BKC. Corresponding heights and hypotenuses therefore satisfy

$$\frac{NK}{BK} = \frac{CT}{BC}. \tag{A.8}$$

Thus

$$CT = \frac{BC \cdot NK}{BK}, \qquad BK \neq 0. \tag{A.9}$$

Replacing CT/BC with its reciprocal would not yield this result. In an isosceles trapezoid of height h and leg length ℓ, the base angle sine is h/ℓ. A segment's vertical component equals its length times that sine. These relations concern the diagram. Interpreting them as physical breadth coordinates still requires the calibrated mapping in Eqs. (2.3) to (2.4).